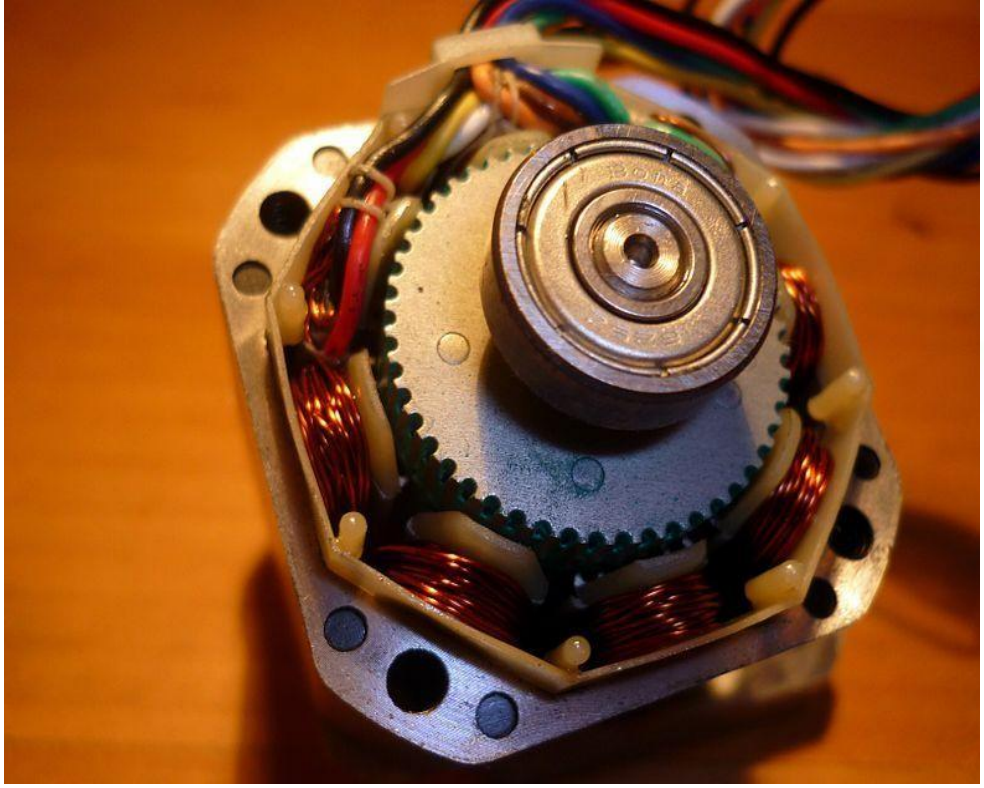




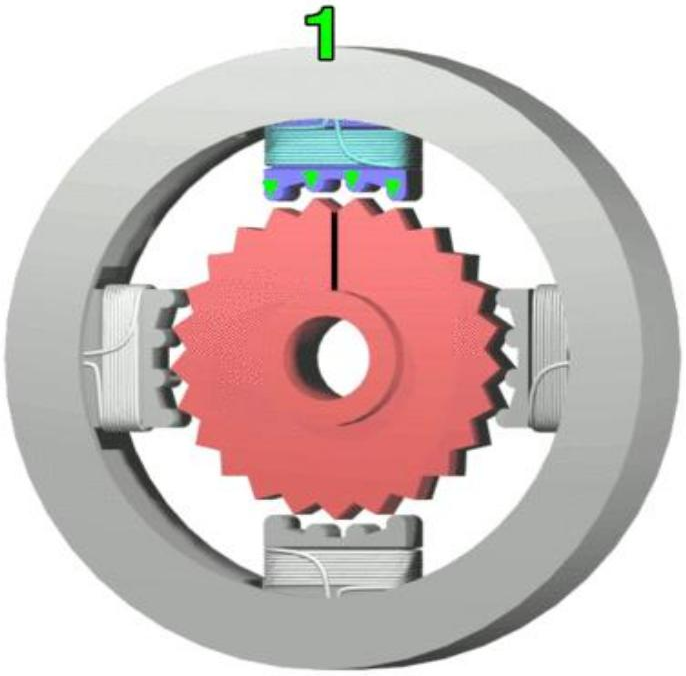
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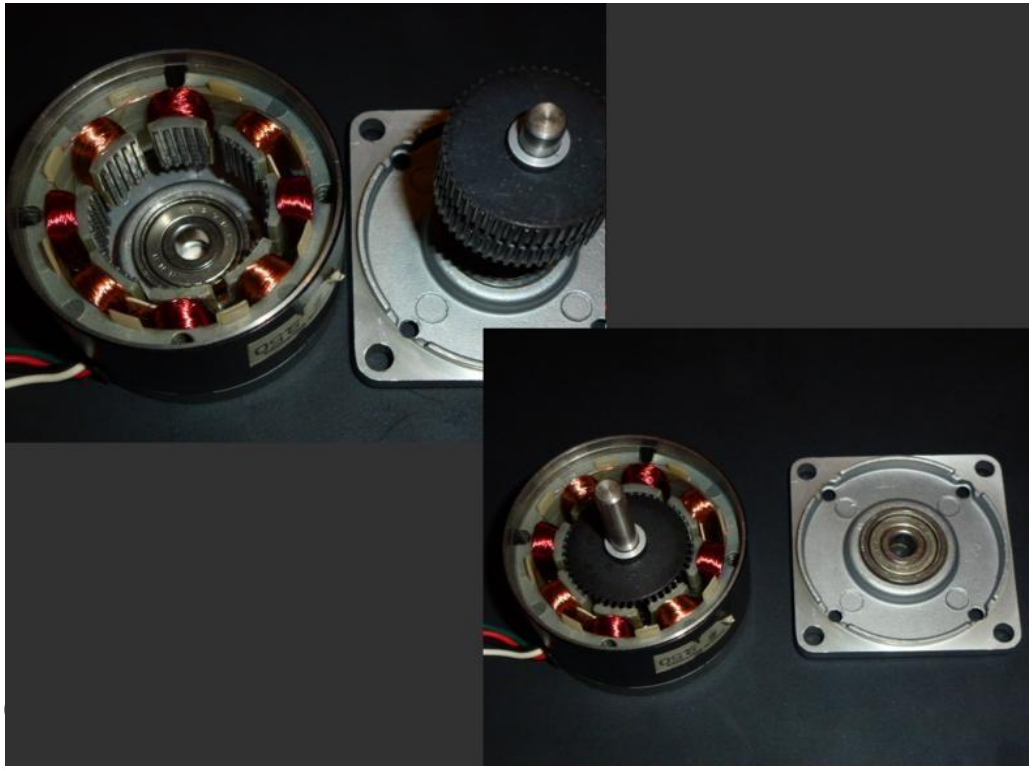
Stepper Motors

Stepper Motor

➢ A **stepper motor** or **step motor** or **stepping motor** is a **brushless DC electric motor** that divides a full rotation into a number of equal steps.

➢The stepper motor is known by its property to convert a train of input pulses (typically square wave pulses) into a precisely defined increment in the shaft position. Each pulse moves the shaft through a fixed angle called **step angle**.





Advantages of stepper motors:

i) They can be directly controlled by computers, micro-processors and programmable controllers.

ii) Precision control of speed without using closed loop feedback.

Applications:

i) **Operation control**:

Such motors used for operational control in computer peripherals, textile industries, IC fabrications and robotics etc.

ii) **Incremental motion**:

Applications requiring incremental motion are typewriters, line printers, tape-drivers, floppy disk drivers, numerically controlled machine tools, process-control systems and X-Y plotters.

Others:

It includes commercial, military and medical applications where these motors perform functions as mixing, cutting, striking, metering, blending and purging. They also take part in manufacture of packed food stuffs, commercial end-products and even the production of science fiction movies.

Step Angle

The angle through which the motor shaft rotates for each command pulse is called the step angle β. Smaller the step angle, greater the number of steps per revolution and higher the resolution or accuracy of positioning obtained. The step angles can be as small as 0.72° or as large as 90°. But the most common step sizes are 1.8°, 2.5°, 7.5° and 15°.

The value of step angle can be expressed either in terms of the rotor and stator poles (teeth) N_r and N_s respectively or in terms of the number of stator phases (m) and the number of rotor teeth.

$$\beta = \frac{(N_s - N_r)}{N_s \cdot N_r} \times 360°$$

or

$$\beta = \frac{360°}{m\,N_r} = \frac{360°}{\text{No. of stator phases} \times \text{No. of rotor teeth}}$$

For example, if $N_s = 8$ and $N_r = 6$, $\beta = (8 - 6) \times 360 / 8 \times 6 = 15°$

Resolution is given by the number of steps needed to complete one revolution of the rotor shaft. Higher the resolution, greater the accuracy of positioning of objects by the motor

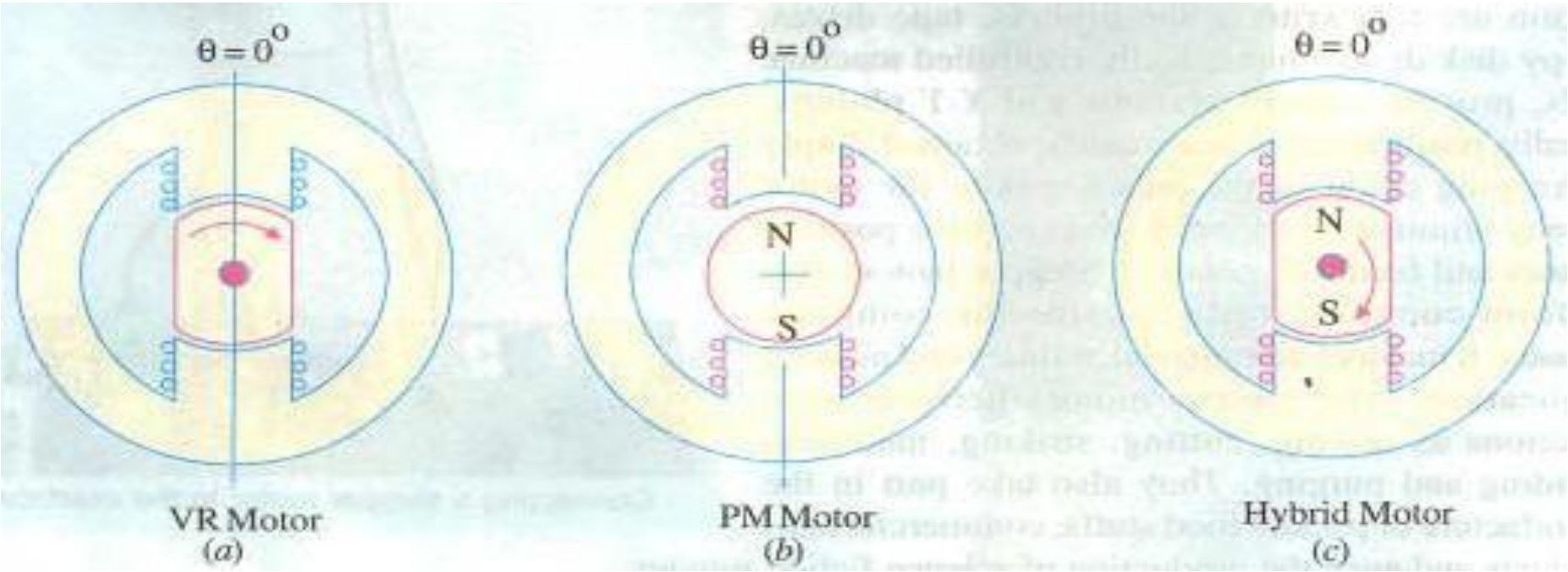
$\therefore$ Resolution = No. of steps / revolution = 360° / β

A stepping motor has the extraordinary ability to operate at very high stepping rates (upto 20,000 steps per second in some motors) and yet to remain fully in synchronism with the command pulses. When the pulse rate is high, the shaft rotation seems continuous. Operation at high speeds is called 'slewing'. When in the slewing range, the motor generally emits an audible whine having a fundamental frequency equal to the stepping rate. If f is the stepping frequency (or pulse rate) in pulses per second (pps) and β is the step angle, then motor shaft speed is given by

$$n = \beta \times f / 360 \text{ rps} = \text{pulse frequency resolution}$$

Stepper Motor Types:

There are three basic stepper motor types:
(i) Variable- reluctance (VR) stepper motor,
(ii) Permanent-magnet (PM) stepper motor and
(iii) Hybrid stepper motor…



39.4. Variable Reluctance Stepper Motors

Construction : A variable-reluctance motor is constructed from ferromagnetic material with salient poles as shown in Fig. 39.2. The stator is made from a stack of steel laminations and has six equally-spaced projecting poles (or teeth) each wound with an exciting coil. The rotor which may be solid or laminated has four projecting teeth of the same width as the stator teeth. As seen, there are three independent stator circuits or phases A, B and C and each one can be energised by a direct current pulse from the drive circuit (not shown in the figure).



Variable reluctance motor

A simple circuit arrangement for supplying current to the stator coils in proper sequence is shown in Fig. 39.2 (e). The six stator coils are connected in 2-coil groups to form three separate circuits called phases. Each phase has its own independent switch. Diametrically opposite pairs of stator coils are connected in series such that when one tooth becomes a N-pole, the other one becomes a S-pole. Although shown as mechanical switches in Fig. 39.2 (e), in actual practice, switching of phase currents is done with the help of solid-state control. When there is no current in the stator coils, the rotor is completely free to rotate. Energising one or more stator coils causes the rotor to step forward (or backward) to a position that forms a path of least reluctance with the magnetized stator teeth. The step angle of this three-phase, four rotor teeth motor is $\beta = 360/\ 4 \times 3 = 30°$.

(*a*) 1-phase-ON or Full-step Operation

Fig. 39.2 (*a*) shows the position of the rotor when switch S_1 has been closed for energising phase *A*. A magnetic field with its axis along the stator poles of phase *A* is created. The rotor is therefore, attracted into a position of minimum reluctance with diametrically opposite rotor teeth 1 and 3 lining up with stator teeth 1 and 4 respectively. Closing S_2 and opening S_1 energizes phase *B* causing rotor teeth 2 and 4 to align with stator teeth 3 and 6 respectively as shown in Fig. 39.2 (*b*). The rotor rotates through full-step of 30° in the clockwise (CW) direction. Similarly, when S_3 is closed after opening S_2, phase *C* is energized which causes rotor teeth 1 and 3 to line up with stator teeth 2 and 5 respectively as shown in Fig. 39.2 (*c*). The rotor rotates through an additional angle of 30° in the clockwise (CW) direction. Next if S_3 is opened and S_1 is closed again, the rotor teeth 2 and 4 will align with stator teeth 4 and 1 respectively thereby making the rotor turn through a further angle of 30° as shown in Fig. 39.2 (*d*). By now the total angle turned is 90°. As each switch is closed and the preceding one opened, the rotor each time rotates through an angle of 30°. By repetitively closing the switches in the sequence 1-2-3-1 and thus energizing stator phases in sequence *ABCA* etc., the rotor will rotate clockwise in 30° steps. If the switch sequence is made 3-2-1-3 which makes phase sequence *CBAC* (or *ACB*), the rotor will rotate anticlockwise. This mode of operation is known as 1-phase-ON mode or full-step operation and is the simplest and widely-used way of making the motor step. The stator phase switching truth table is shown in Fig. 39.2 (*f*). It may be noted that the direction of the stator magnetizing current is not significant because a stator pole of either magnetic polarity will always attract the rotor pole by inducing opposite polarity.

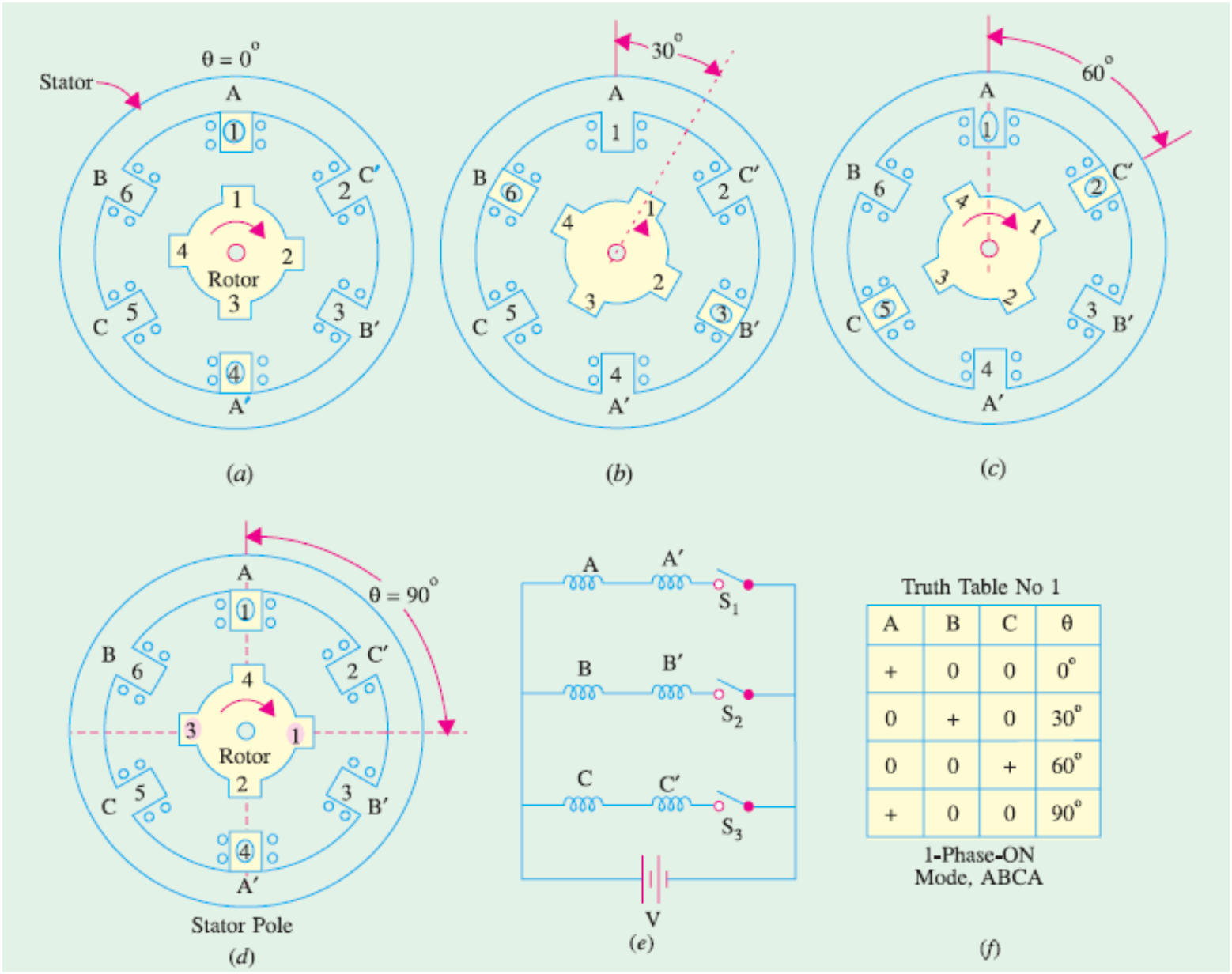

θ = 0°
Stator
A
B
C
C′
B′
A′
Rotor
(a)
30°
(b)
60°
(c)
θ = 90°
Stator Pole
(d)
S1
S2
S3
V
(e)
Truth Table No 1
A | B | C | θ
+ | 0 | 0 | 0°
0 | + | 0 | 30°
0 | 0 | + | 60°
+ | 0 | 0 | 90°
1-Phase-ON
Mode, ABCA
(f)

Fig. 39.2

(*b*) 2-phase-ON Mode

In this mode of operation, two stator phases are excited simultaneously. When phases *A* and *B* are energized together, the rotor experiences torques from both phases and comes to rest at a point mid-way between the two adjacent full-step positions. If the stator phases are switched in the sequence *AB*, *BC*, *CA*, *AB* etc., the motor will take full steps of 30° each (as in the 1-phase-*ON* mode) but its equilibrium positions will be interleaved between the full-step positions. The phase switching truth table for this mode is shown in Fig. 39.3 (*a*).

Truth Table No. 2

A	B	C	
+	+	0	15°
0	+	+	45°
+	0	+	75°
+	+	0	105°

2 Phase-ON Mode
AB, BC, CA, AB

Truth Table No. 3

	A	B	C		
A	+	0	0	0°	
	+	+	0	15°	AB
B	0	+	0	30°	
	0	+	+	45°	BC
C	0	0	+	65°	
	+	0	+	75°	CA
A	+	0	0	90°	

Half-Stepping Alternate
1-Phase-On &
2-Phase-on Mode
A, AB, B, BC, C, CA, A

Fig. 39.3

The 2-phase-ON mode provides greater holding torque and a much better damped single-stack response than the 1-phase-ON mode of operation.

(c) Half-step Operation

Half-step operation or 'half-stepping' can be obtained by exciting the three phases in the sequence A, AB, B, BC, C etc. *i.e.* alternately in the 1-phase-ON and 2-phase-ON modes. It is sometime known as 'wave' excitation and it causes the rotor to advance in steps of 15° *i.e.* half the full-step angle. The truth table for the phase pulsing sequence in half-stepping is shown in Fig. 39.3 (*b*).

Half-stepping can be illustrated with the help of Fig. 39.4 where only three successive pulses have been considered. Energizing only phase A causes the rotor position shown in Fig. 39.4 (*a*). Energising phases A and B simultaneously moves the rotor to the position shown in Fig. 39.4 (*b*) where rotor has moved through half a step only. Energising only phase B moves the rotor through another half-step as shown in Fig. 39.4 (*c*). With each pulse, the rotor moves $30 / 2 = 15°$ in the CCW direction.

It will be seen that in half-stepping mode, the step angle is halved thereby doubling the resolution. Moreover, continuous half-stepping produces a smoother shaft rotation.

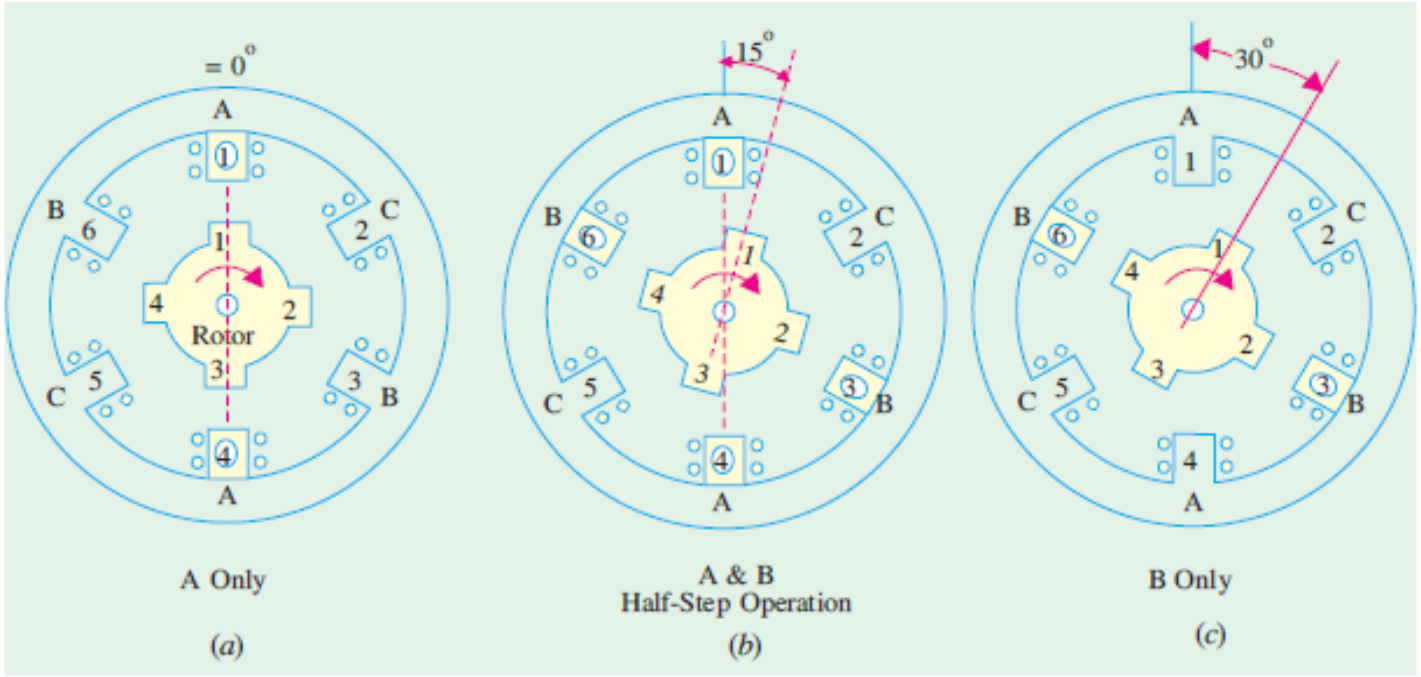


Fig. 39.4

What is micro-stepping?

It is also known as mini-stepping. It utilizes two phases simultaneously as in 2-phase-*ON* mode but with the two currents deliberately made unequal (unlike in half-stepping where the two phase currents have to be kept equal). The current in phase *A* is held constant while that in phase *B* is increased in very small increments until maximum current is reached. The current in phase *A* is then reduced to zero using the same very small increments. In this way, the resultant step becomes very small and is called a microstep. For example, a *VR* stepper motor with a resolution of 200 steps / rev ($\beta = 1.8°$) can with microstepping have a resolution of 20,000 steps / rev ($\beta = 0.018°$). Stepper motors employing microstepping technique are used in printing and phototypesetting where very fine resolution is called for. As seen, microstepping provides smooth low-speed operation and high resolution.

Permanent Magnet Stepper motor:

It has wound stator poles and permanently magnetized rotor poles. It has a cylindrical rotor. Its direction of rotation depends on the polarity of stator current. The stator has projecting poles. Rotor has 2 poles whereas stator has 4 poles and the two stator poles are energized by one winding, the motor has two windings or phases.

39.6. Permanent-Magnet Stepping Motor

(*a*) **Construction.** Its stator construction is similar to that of the single-stack *VR* motor discussed above but the rotor is made of a permanent-magnet material like magnetically 'hard' ferrite. As shown in the Fig. 39.6 (*a*), the stator has projecting poles but the rotor is cylindrical and hasradially magnetized permanent magnets. The operating principle of such a motor can be understood with the help of Fig. 39.6 (*a*) where the rotor has two poles and the stator has four poles. Since two stator poles are energized by one winding, the motor has two windings or phases marked *A* and *B*. The step angle of this motor $\beta = 360° / mN_r = 360° / 2 \times 2 = 90°$ or $\beta = (4 - 2) \times 360° / 2 \times 4 = 90°$.



Permanent magnet stepper motor

Working :

When a particular stator phase is energized, the rotor magnetic poles move into alignment with excited stator poles. The stator wdgs. A and B can be excited with either polarity current (A^+ refers to the +ve current i^+_A in phase A and A^- to negative current (i^-_A). Fig, shows condition when phase A is excited with the current i^+_A. Here $\theta = 0^o$. If excitation is now switched to phase B, the rotor rotates by a full step of 90° clockwise direction. When phase A is excited with –ve current i^-_A, the rotor turns through another 90° in clockwise direction. Similarly excitation of phase B with i^+_B further turns rotor through another 90° in same direction. After this excitation of phase A with i^+_A makes rotor run through one complete revolution of 360°.

$i^+_A;\ i^+_B;\ i^-_A,\ i^-_B, i^+_A$

$A^+; B^+; A^-; B^-; A^+$ } Clockwise

$i^+_A;\ i^-_B;\ i^-_A,\ i^+_B, i^+_A$

$A^+; B^-; A^-; B^+; A^+$ } CCW

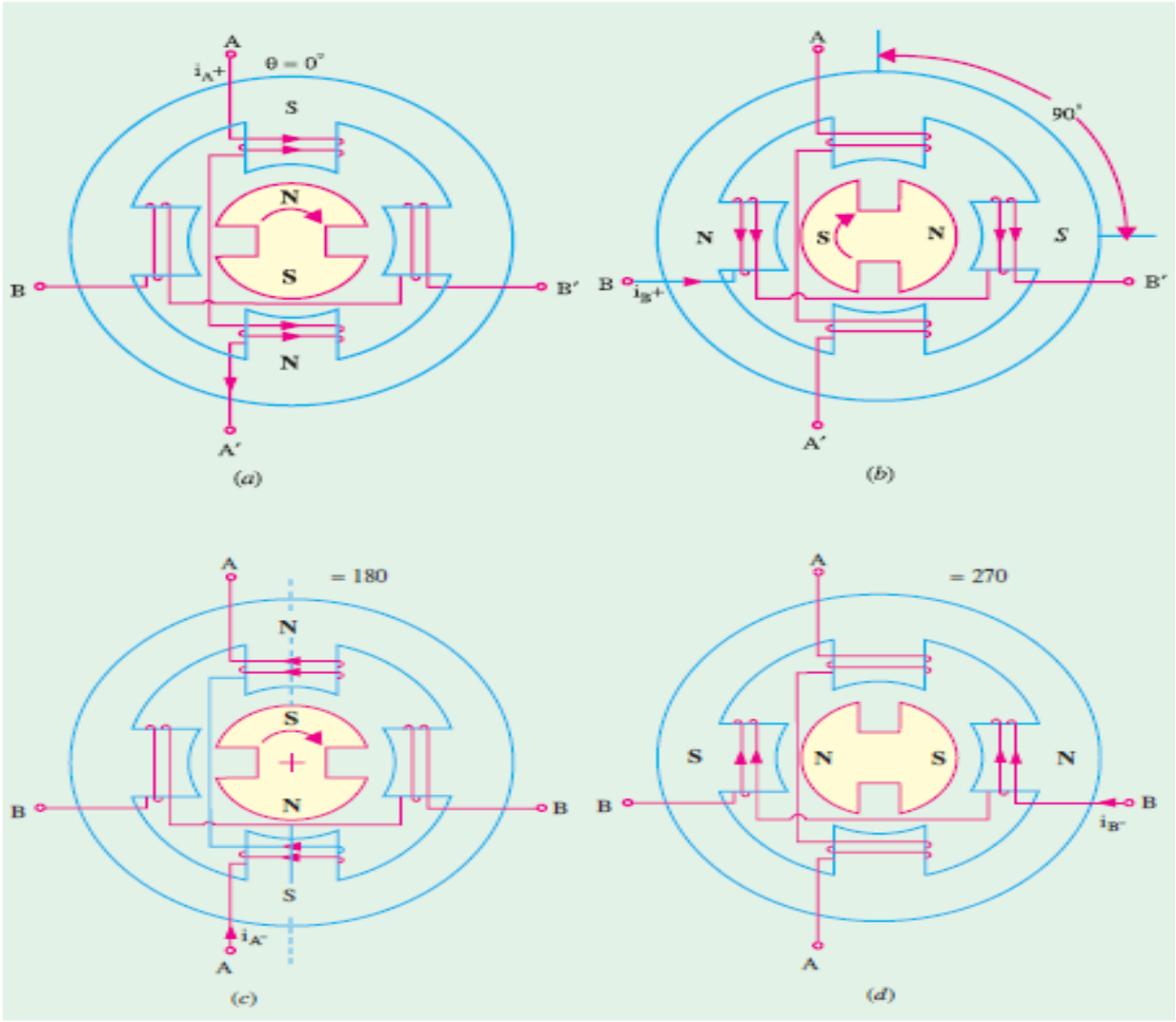

A
i_{A+}
θ = 0°
S
N
B
B′
S
N
A′
(a)
A
90°
N
S
N
S
B
i_{B+}
B′
A′
(b)
A
= 180
N
S
+
N
B
B
S
i_{A-}
A
(c)
A
= 270
S
N
S
N
B
B
i_{B-}
A
(d)

Fig. 39.6


THANK YOU